JAMA | Review

IgA Nephropathy in Adults
A Review

Sinead Stoneman, MB, BCh, BAO; Jia Wei Teh, MB, BCh, BAO, MSc;
Michelle Marie O'Shaughnessy, MB, BCh, BAO, MD, MS

IMPORTANCE IgA nephropathy (IgAN) is a chronic kidney disease involving deposition of IgA-containing immune complexes in the glomerulus, causing glomerular inflammation and scarring. It is the most common immune-mediated glomerular disease worldwide, and affects an estimated 198 887 to 208 184 persons in the US. Up to 50% of patients with IgAN develop kidney failure within 10 years of diagnosis.

OBSERVATIONS IgAN typically presents with nephritic syndrome and usually occurs in younger adults, with a mean age at diagnosis of 34 to 45 years. Incidence is highest in East Asia. Approximately 60% of cases are detected incidentally with hematuria or proteinuria on urinalysis. Up to 30% of patients present with episodic visible hematuria, often concomitantly with an upper respiratory or gastrointestinal tract infection (synpharyngitic hematuria). Less common presentations include nephrotic syndrome (<5%) and rapidly progressive glomerulonephritis (<5%). When IgAN is suspected (due to hematuria, proteinuria, or reduced kidney function), initial workup should include quantification of proteinuria and assessment for other causes of nephritic syndrome (eg, lupus nephritis). Adults with suspected IgAN and proteinuria greater than or equal to 0.5 g per day should undergo kidney biopsy. The diagnosis of primary IgAN is based on presence of IgA-dominant immune deposits in the glomerular mesangium after excluding other causes of this histologic appearance, ie, IgA vasculitis, IgA-dominant infection-related glomerulonephritis, and secondary IgAN from diseases such as cirrhosis, inflammatory bowel disease, celiac disease, infection (eg, viral hepatitis), and autoimmune diseases (eg, axial spondyloarthritis). Based on the Kidney Disease: Improving Global Outcomes 2025 clinical practice guideline for the management of IgAN, treatment for patients with proteinuria greater than 0.5 g per day includes behavioral modifications (eg, dietary sodium <2 g/d, smoking cessation, weight control, exercise), antihypertensive medications for goal blood pressure less than 120/70 mm Hg, and therapies to reduce the formation of IgA-containing immune complexes (eg, targeted-release budesonide), decrease glomerular injury (eg, systemic glucocorticoids, iptacopan), and manage existing IgAN-induced nephron loss (eg, renin-angiotensin system inhibitor or dual endothelin angiotensin receptor antagonist [eg, sparsentan] alone or in combination with a sodium-glucose cotransporter 2 inhibitor).

CONCLUSIONS AND RELEVANCE IgAN is the leading cause of immune-mediated glomerular disease worldwide. Patients with suspected IgAN and proteinuria greater than or equal to 0.5 g per day should undergo kidney biopsy to confirm the diagnosis. Treatment of IgAN includes behavioral modifications, blood pressure management, and therapies to decrease formation of IgA-containing immune complexes (eg, targeted-release budesonide), reduce immune complex–mediated glomerular injury (eg, systemic glucocorticoids, iptacopan), and manage IgAN-induced nephron loss (eg, renin-angiotensin system inhibitor, dual endothelin angiotensin receptor antagonist, and sodium-glucose cotransporter 2 inhibitor).

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Author Affiliations: Division of Nephrology, University of British Columbia, Vancouver, British Columbia, Canada (Stoneman); Department of Nephrology, University Hospital Galway, Galway, Ireland (Teh, O'Shaughnessy); School of Medicine, University of Galway, Galway, Ireland (O'Shaughnessy).

Corresponding Author: Michelle Marie O'Shaughnessy, MB, BCh, BAO, MD, MS, Department of Nephrology, Galway University Hospitals, Galway H91TY80, Ireland (m.oshaughnessy4@hse.ie).

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IgA nephropathy (IgAN) involves deposition of IgA-containing immune complexes in the glomerular mesangium, triggering glomerular inflammation and scarring.[1] It is the most common cause of immune-mediated glomerular disease worldwide, with an estimated global incidence between 1.4 and 2.5 per 100 000 persons in the US,[2,3] although incidence is likely underreported because milder disease may not be detected. Up to 50% of patients with IgAN develop kidney failure within 10 years of diagnosis,[4] and life expectancy is estimated to be 6 years shorter for those with IgAN compared with matched controls.[5]

Peak IgAN incidence occurs at a mean age of 34 to 45 years,[4,6-9] and is highest in persons of Asian and European ancestry.[10] Approximately 60% of cases are detected when hematuria, proteinuria, or impaired kidney function are incidentally identified.[11] Up to 30% of patients present with episodic visible hematuria, often concomitantly with an upper respiratory or gastrointestinal tract infection, ie, synpharyngitic hematuria.[12] Primary IgAN is diagnosed when kidney biopsy shows IgA-dominant immune deposits in the glomerular mesangium and other causes of IgA-dominant glomerulonephritis (ie, IgA vasculitis, IgA-dominant infection-related glomerulonephritis, secondary cause of IgAN such as chronic liver disease) are excluded.[13]

This article summarizes the pathophysiology, epidemiology, clinical presentation, diagnosis, treatment, and prognosis of primary IgAN in adults, with a focus on novel therapies recently approved for IgAN and treatment recommendations based on the 2025 Kidney Disease: Improving Global Outcomes (KDIGO) clinical practice guidelines for IgAN.

Methods

A PubMed database search was performed for English-language studies of adults using the expanded medical subject headings term *glomerulonephritis IGA*, retrieving 7088 articles published between January 1, 1954, and November 17, 2025 (eTable 1 in the Supplement). We preferentially reviewed and cited articles based on recency of publication, relevance, and quality of evidence. References from selected articles were manually reviewed to identify additional articles of potential relevance. This review includes 97 articles: 4 clinical practice guidelines, 1 Cochrane review, 3 meta-analyses, 2 systematic reviews, 19 randomized clinical trials (RCTs), 2 prospective cohort studies, 34 retrospective cohort studies, 18 case-control studies, 1 descriptive epidemiologic study, 1 case series, 1 animal study, 1 genome-wide association study, and 10 narrative reviews.

Discussion/Observations

Pathophysiology of IgAN

Four consecutive steps, termed the *4-hit hypothesis* must occur for primary IgAN to develop (**Figure 1**). First, patients develop aberrant production of galactose-deficient IgA1 (Gd-IgA1) immunoglobulins at the gut or respiratory mucosa, which may be triggered by gastrointestinal or respiratory tract infection[14-17] or by an altered gut microbiome.[18] The production of aberrant Gd-IgA1 persists even after the triggering insult resolves. Second, Gd-IgA1 molecules must travel to the systemic circulation, where they are recognized by circulating IgG and IgA, resulting in the third step of Gd-IgA1–containing immune complex formation.[1] In the fourth step, Gd-IgA1–containing immune complexes deposit in the glomerular mesangium, triggering local inflammation, complement activation, mesangial cell proliferation, and podocyte and tubulointerstitial injury.[19] All 4 steps appear necessary for clinical disease to manifest; aberrant production of Gd-IgA1 was identified in up to 25% of unaffected family members of patients with IgAN[20] and glomerular IgA deposition was identified in 16% of healthy kidney donors in Japan.[21] As with other forms of chronic kidney disease (CKD), once IgAN-induced nephron loss has occurred, the remaining nephrons are exposed to glomerular hypertension and hyperfiltration, resulting in further nephron loss, tubulointerstitial fibrosis, and clinical CKD progression.

It is unclear why IgAN is a kidney-limited disease but IgA vasculitis (formerly *Henoch Schönlein purpura*)—which shares similar pathogenic mechanisms and histologic features with IgAN—also affects the skin, joints, and gastrointestinal tract.[22,23] Impaired clearance of IgA-containing immune complexes in chronic liver disease[24] and formation of antigliadin IgA antibodies in celiac disease[25] may contribute to the development of secondary IgAN.

Epidemiology of IgAN

IgAN typically presents in younger adults, with a mean age at diagnosis of 34 to 45 years, and is more common among males than females (with a 2:1 ratio among White individuals).[4,6-8] Demographic and clinical characteristics of large cohorts of patients with IgAN are shown in **Table 1** and eTable 2 in the Supplement.

The reported incidence of primary IgAN varies across regions due to differences in true disease incidence, urinary screening practices (screening of school-aged children is practiced in some countries), and access to diagnostic kidney biopsy. Worldwide, primary IgAN incidence is likely to be underreported because mild cases may not become clinically apparent and those with mild or advanced kidney disease may not undergo kidney biopsy. Nevertheless, incidence is estimated at 0.76 per 100 000 per year in Europe,[26] 1.41 per 100 000 per year in Queensland, Australia,[27] 1.8 per 100 000 per year in Singapore,[28] and 1.40 per 100 000 per year in Southern California.[29] Based on data from a nationwide survey published in 2025 of 43 US pathologists who estimated the proportion of kidney biopsies showing IgAN in their practice, the annual incidence of IgAN in the US was extrapolated to 1.9 to 2.2 per 100 000, with an estimated prevalence of 54 to 63 per 100 000 and a total US prevalence of 198 887 to 208 184 persons.[30]

IgAN is more common among individuals with Asian ancestry and less frequent in those with African ancestry. In the International Kidney Biopsy Survey, which collected kidney biopsy data from 29 international centers (2012-2013), 42 603 glomerular disease diagnoses were reported. IgAN accounted for 40% of cases in Japan and Thailand, 22% in Europe (13 centers across 10 countries), and 12% in North America (10 centers across the US and Canada).[10] In contrast, IgAN was rare (5.8% of all biopsy diagnoses) in a multiracial cohort of 1284 patients who underwent kidney biopsy in South Africa (2000-2009).[31] Within an integrated health system in Southern California, biopsy-proven IgAN incidence (2010-2021) was 4.5 per 100 000 in Asian or Pacific Islander, 1.7 per 100 000 in Hispanic

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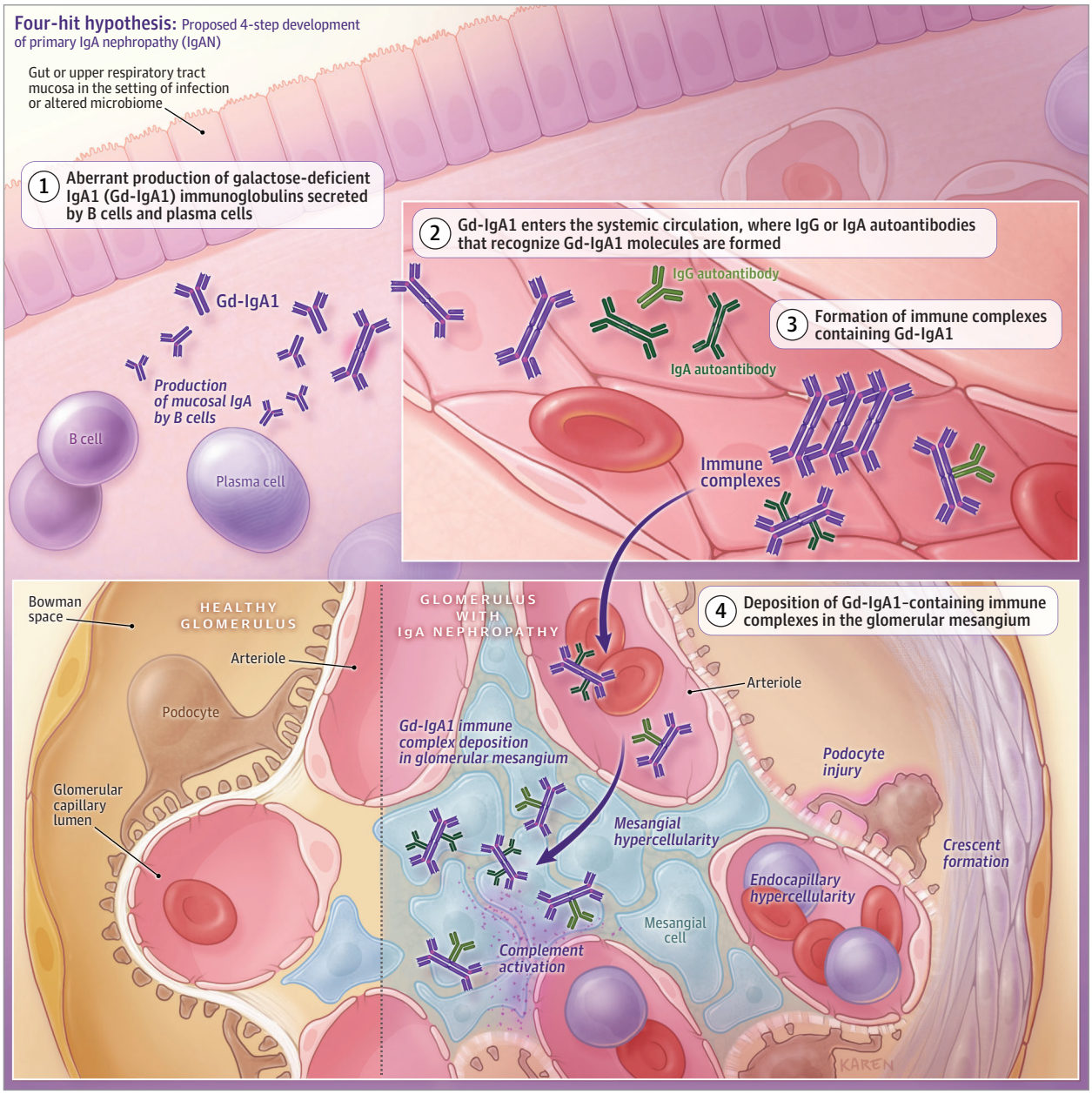
Figure 1. Pathophysiology of IgA Nephropathy



or Latino, 1.2 per 100 000 in White, and 0.6 per 100 000 in Black participants.[29]

Risk Factors for IgAN Development

Several genetic and environmental risk factors for IgAN have been identified. The largest meta genome-wide association study to date, which included 10 146 IgAN cases and 28 751 matched controls, identified 30 risk loci that explained 11% of overall IgAN risk.[2] Among nongenetic risk factors, recurrent episodes of tonsillitis are associated with an increased risk of developing IgAN. Among 4 311 393 adults in Japan without a preceding IgAN diagnosis, the incidence of IgAN was 4.93 (95% CI, 3.24-7.48) per 10 000 person-years in those with chronic tonsillitis determined via *International Statistical Classification of Diseases and Related Health Problems, Tenth Revision* code, compared with 1.74 (95% CI, 1.68-1.81) per 10 000 person-years in those without chronic tonsillitis (adjusted hazard ratio [HR], 2.72 [95% CI, 1.79-4.14]).[16]

Clinical Presentation

Most patients with IgAN present with some or all elements of the nephritic syndrome, ie, nonvisible hematuria, proteinuria, impaired kidney function, and hypertension (**Figure 2** and **Box**). Reported relative frequencies of discrete presentations within this spectrum are highly variable due to differences in study design (geographic region, time period, prospective or retrospective data collection, case definitions) and diagnostic pathways. Countries with school urinary screening programs (eg, Japan, Taiwan, South Korea) detect cases that are milder or earlier in their disease

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Table 1. An Overview of the Epidemiology of IgAN From Select Large International Cohort Studies

Characteristic	RaDaR[4] (N = 2439)	VALIGA[6] (N = 1147)	Toronto[7] (N = 542)	China[8] (N = 1155)	Southern California[9] (N = 655)
Region or country	UK	Europe[a]	Canada	China	Southern California, US
Population	Adults and children; 94.3% adults ≥18 y	Adults and children; 85% adults ≥18 y	100% Adults a nd older adolescents ≥16 y	100% Adults ≥18 y	100% Adults ≥18 y
Baseline characteristics					
Age at diagnosis, mean (SD), y	41 (15)	36 (16)	38 (13)	34 (9)	45 (15)
Sex, No. (%)					
Female	716 (29)	NA (27)	NA (39)	NA (50)	311 (48)
Male	1723 (71)	NA (73)	NA (61)	NA (50)	344 (53)
Race and ethnicity, No. (%)					
Asian or Pacific Islander	228 (9)	NA (<1)	NA (23)		201 (31)
Black	32 (1)	NA (<1)	NA (3)		20 (3)
Hispanic/Latino					259 (40)
White	1885 (77)	NA (98)	NA (50)		159 (24)
Multiple	50 (2)	NA (2)	NA (10)		
Not stated/missing	244 (10)		NA (14)		16 (2)
Proteinuria at diagnosis					
Mean (SD) or median (IQR), g/d		1.3 (0.6 to 2.6)[b]	2.4 (2.5)[c]	0.9 (0.5 to 1.6)[b]	
uPCR at diagnosis, mean (SD), g/g	2.4 (3.6)				2.5 (2.4)[d]
eGFR at diagnosis, mean (SD), mL/min/1.73 m^2	55 (29)	73 (30)	77 (33)[e]	89 (33)	60 (30)
Kidney outcomes					
Length of follow-up, mean (SD) or median (IQR), y	5.9 (3.0 to 10.5)	4.7 (2.4 to 7.9)	6.5 (4.9)	5.4 (4.1 to 7.2)	3.3 (1.5 to 7.0)
Kidney failure, No. (%)	1210 (50)[f]	NA (12)[g]	NA (28)[h]	108 (9)[i]	179 (27)[i]
Rate of eGFR/CrCl change					
Mean (SD) or median (IQR), mL/min/1.73 m^2/y	−3.6 (8.2)	−1.8 (7.5)		−1.7 (−4.6 to −1.0)	
Mean (SD), mL/min/1.73 m^2/mo			−0.4 (0.6)		
Estimated 10-y kidney survival, % (95% CI)	54 (51 to 56)[j]	74[k]		83 (79 to 87)[k]	

Abbreviations: CrCl, creatinine clearance; eGFR, estimated glomerular filtration rate; NA, not available; uPCR, urine protein to creatinine ratio.

[a] Czech Republic, Estonia, Germany, Greece, Italy, Poland, Portugal, Sweden, Spain, Netherlands, Turkey, and the UK.

[b] Proteinuria at time of biopsy.

[c] Proteinuria at time of diagnosis.

[d] uPCR at baseline.

[e] CrCl.

[f] Kidney failure or death event: death = 21 (2%); dialysis = 298 (25%); transplant = 95 (8%); eGFR <15 mL/min/1.73 m^2 = 796 (66%). Mean (SD) time to first event, 6.9 (7.1) years. Mean (SD) age at first event, 48 (15) years.

[g] Kidney failure defined as eGFR <15 mL/min/1.73 m^2.

[h] Kidney failure defined as CrCl <15 mL/min/1.73 m^2, initiation of dialysis, or transplant.

[i] Kidney failure defined as eGFR <15 mL/min/1.73 m^2, initiation of dialysis, or transplant.

[j] Survival rate defined by absence of kidney failure and death.

[k] Survival rate defined by absence of combined end point of 50% reduction in eGFR or kidney failure.

course, whereas countries with limited access to diagnostic kidney biopsy (eg, low- and middle-income countries) have more severe or advanced presentations.

Approximately 10% to 30% of patients with IgAN have visible hematuria, often concomitantly with an upper respiratory or, less commonly, gastrointestinal infection, ie, synpharyngitic hematuria.[32] Less than 10% of patients with IgAN present with more rapid loss of kidney function due to severe glomerular inflammation (ie, rapidly progressive glomerulonephritis), malignant hypertension with superimposed thrombotic microangiopathy,[33] or acute kidney injury (AKI) in the setting of visible hematuria causing kidney tubular injury.[12] Less than 5% of patients with IgAN present with the nephrotic syndrome (ie, edema, hypoalbuminemia, and >3.5 g/d urine protein), which may represent background IgAN with superimposed minimal change disease.

Differential Diagnosis

The differential diagnosis for IgAN initially includes any cause of proteinuria, hematuria, or reduced kidney function, including urologic conditions (eg, cystitis, bladder tumor), which typically cause hematuria without proteinuria; common causes of CKD (eg, diabetic nephropathy); and other rarer causes of glomerular disease (eg, postinfectious glomerulonephritis, antineutrophil cytoplasmic antibody–associated glomerulonephritis, lupus nephritis, Alport syndrome [a genetic disorder involving abnormal production of type IV collagen]).

Among patients with kidney biopsy results showing IgA-dominant glomerulonephritis, the differential diagnosis includes IgA vasculitis, IgA-dominant infection-related glomerulonephritis, and secondary forms of IgAN, which are all histologically indistinguishable from primary IgAN. IgA vasculitis typically presents with

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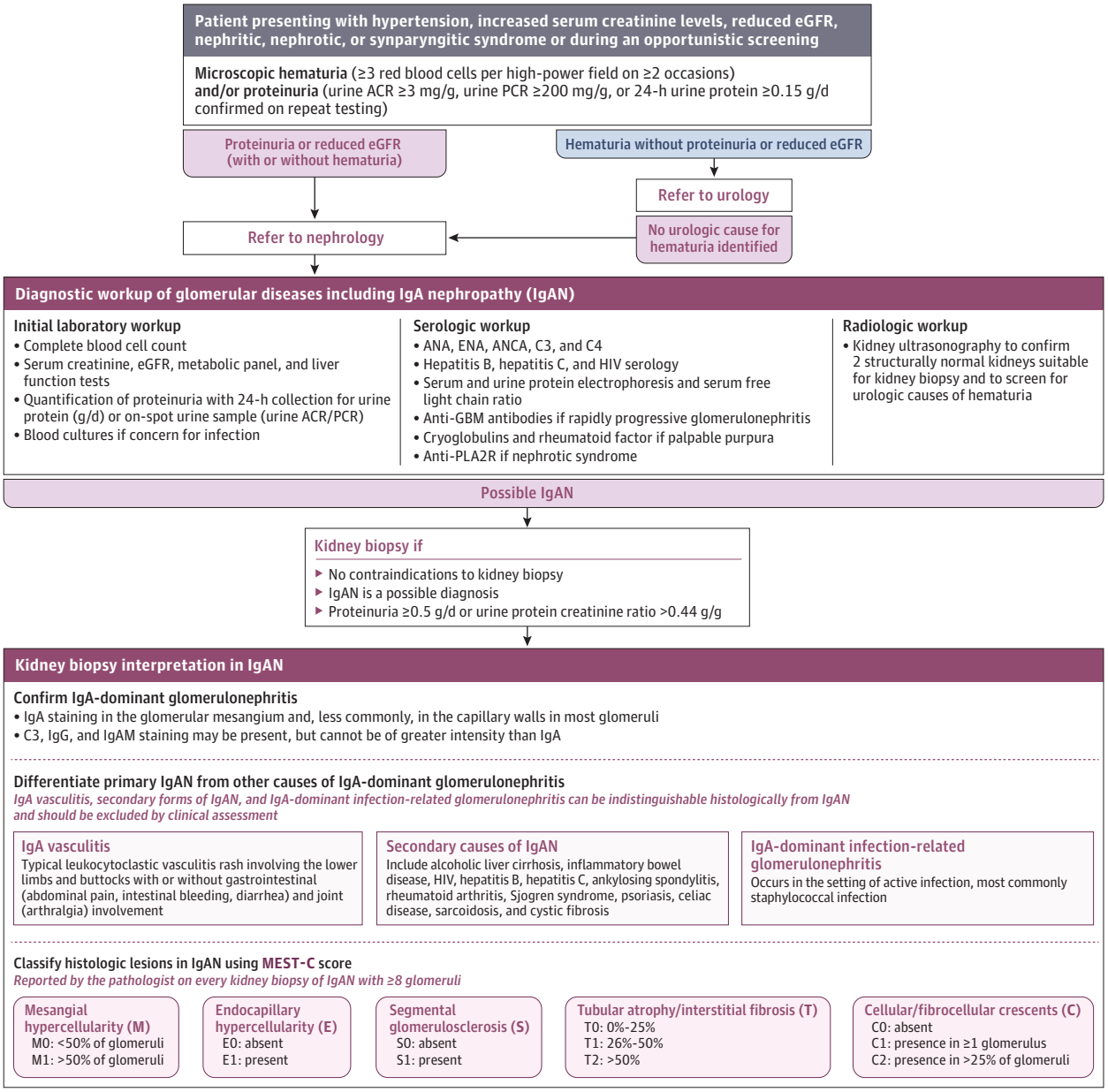
Figure 2. Initial Evaluation of Suspected IgAN and Kidney Biopsy Interpretation in IgAN



ACR indicates albumin to creatinine ratio; ANA, antinuclear antibody; ANCA, antineutrophil cytoplasmic antibodies; eGFR, estimated glomerular filtration rate; ENA, extractable nuclear antigen test; GBM, glomerular basement membrane; PCR, protein to creatinine ratio; PLA2R, phospholipase A2 receptor.

a vasculitic rash (palpable purpura), usually on the buttocks and lower limbs; arthralgias or arthritis, especially of the knees and ankles; and abdominal pain with or without lower gastrointestinal bleeding.[34] IgA-dominant infection-related glomerulonephritis, first described among patients with diabetes and *Staphylococcus aureus* soft tissue infections,[35] can occur with other bacterial infections such as *Escherichia coli* and *Streptococcus* species.

Secondary IgAN

More than 50 diseases have been reported in association with IgAN, including gastrointestinal (eg, cirrhosis, inflammatory bowel disease, celiac disease), autoimmune (eg, rheumatoid arthritis, axial spondyloarthritis), infectious (viral hepatitis, HIV), and dermatologic (eg, psoriasis or dermatitis herpetiformis—an autoimmune blistering skin disease) diseases.[36] However, a consensus definition of secondary IgAN is lacking,[37] and the number of reported cases for associated diseases was typically less than 10 in a systematic review that analyzed 72 case reports of secondary IgAN performed in 2022, precluding any definitive conclusions regarding causality, natural history, or treatment effectiveness.[38] In a retrospective analysis of insurance claims data from more than 4 million adults in Japan, the incidence rate of IgAN was 3.43 (95% CI, 2.26-5.21) per 10 000 person-years in those with inflammatory bowel disease compared with 1.74 (95% CI, 1.68-1.81) per 10 000 person-years in those

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Box. Commonly Asked Questions About IgA Nephropathy (IgAN)

How does IgAN typically present?
IgAN usually presents in younger adults (35-45 years), with a 2:1 male predominance in White populations. Most patients have some or all elements of the nephritic syndrome (ie, nonvisible hematuria, proteinuria, and impaired kidney function), but 10% to 30% have episodes of visible hematuria, often concomitantly with an upper respiratory or gastrointestinal infection (ie, synpharyngitic hematuria). Less than 10% of patients with IgAN present with more rapid loss of kidney function due to severe glomerular inflammation or tubular toxicity from hematuria, and less than 5% of patients present with the nephrotic syndrome.

How is IgAN diagnosed?
Kidney biopsy, which is required to confirm a diagnosis of IgAN, should be performed in all adults with suspected IgAN and urine protein excretion greater than 0.5 g/d (or a urine protein creatinine ratio >0.44 g/g). If kidney biopsy reveals IgA-dominant glomerulonephritis, patients should be assessed for signs and symptoms of IgA vasculitis, infections such as *Staphylococcus aureus* soft tissue infections that can cause IgA-dominant infection–related glomerulonephritis, and diseases associated with secondary forms of IgAN (eg, cirrhosis, inflammatory bowel disease, celiac disease, rheumatoid arthritis, axial spondyloarthritis, HIV). Primary IgAN is a diagnosis of exclusion after these diagnoses have been excluded.

How should primary IgAN be managed?
Based on the Kidney Disease: Improving Global Outcomes 2025 clinical practice guidelines for IgAN, treatment includes behavioral modifications to reduce kidney and cardiovascular risk (dietary sodium <2 g/d, smoking cessation, weight control, exercise) and antihypertensive medications with a goal blood pressure less than 120/70 mm Hg. For patients with proteinuria greater than 0.5 g per day, therapies include medications to reduce formation of IgA-containing immune complexes (eg, targeted-release budesonide), decrease glomerular injury (eg, systemic glucocorticoids, iptacopan), and manage existing IgAN-induced nephron loss (eg, renin-angiotensin system inhibition or sparsentan, singly or in combination with sodium-glucose cotransporter 2 inhibition).

without inflammatory bowel disease (adjusted HR, 1.96 [95% CI, 1.29-2.99]).[39] In a retrospective analysis of 1 224 955 adults in the UK, the incidence of IgAN was 7.73 (95% CI, 3.05-17.62) per 100 000 person-years in those with moderate to severe psoriasis compared with 1.71 (95% CI, 1.42-2.06) per 100 000 person-years in those without psoriasis (adjusted HR, 4.75 [95% CI, 1.92-11.76]).[40] Treatment of secondary IgAN focuses on treating the underlying disease, along with general management of CKD.[37]

Assessment and Diagnosis

For patients with suspected IgAN, clinicians should assess for episodic visible hematuria, frothy urine (signifying proteinuria), and peripheral edema and inquire about a family history of IgAN or CKD.[41] Physical examination findings are normal in most patients with IgAN, other than hypertension or, less commonly, peripheral edema.

Laboratory testing should include creatinine (and/or cystatin C) with calculation of estimated glomerular filtration rate (eGFR), urine protein (spot albumin to creatinine ratio or protein to creatinine ratio as initial screening, usually with subsequent confirmatory 24-hour urine protein measurement), and testing for hematuria (urine dipstick initially, ideally with subsequent urine microscopy to identify dysmorphic red blood cells or red blood cell casts). Additional laboratory tests that identify other diagnoses, and potentially obviate the need for kidney biopsy, include hemoglobin A_{1C}, antinuclear antibody, complement C3/C4, antineutrophil cytoplasmic antibodies, antiglomerular basement membrane antibody, HIV/hepatitis B/ hepatitis C serology, serum protein electrophoresis with serum free light chain ratio, and serum antiphospholipase A2 receptor antibodies with or without screening for additional membranous nephropathy-related antibodies.

Although radiologic testing is not used to diagnose IgAN, kidney ultrasonography or abdominal computed tomographic urography without and with intravenous contrast should be considered for patients with visible hematuria to exclude a urologic source of bleeding, especially in those without proteinuria. In addition, kidney ultrasonography is typically performed before kidney biopsy to confirm the presence of 2 structurally normal kidneys.

Kidney biopsy, typically performed by interventional radiologists or nephrologists, is required to make a diagnosis of IgAN. The KDIGO 2025 clinical practice guideline for IgAN recommends kidney biopsy in patients with suspected IgAN (ie, unexplained hematuria, proteinuria, and/or reduced kidney function) and proteinuria exceeding 0.5 g/d or a urine protein to creatinine ratio of approximately 0.44 g/g[13] (Figure 2; eTable 3 in the Supplement). Characteristic light microscopy findings of IgAN include focal or diffuse mesangial proliferation and expansion of the extracellular matrix with or without crescents in the Bowman space. Immunofluorescence microscopy and electron microscopy show IgA deposition in the glomerular mesangium and, less commonly, in the capillary walls.[42]

After IgA-dominant glomerulonephritis is confirmed, patients should be reassessed for symptoms or signs of IgA vasculitis, active infection (causing IgA-dominant infection-related glomerulonephritis), and diseases associated with secondary IgAN, because these conditions are histologically indistinguishable from primary IgAN (Figure 2). Primary IgAN is a diagnosis of exclusion, considered only after these differential diagnoses have been excluded.

Risk Factors for Progression of IgAN

Many modifiable and nonmodifiable demographic, clinical, and histological factors are associated with risk of kidney failure in IgAN (**Table 2**).[43-51] Among clinical factors, proteinuria is the strongest predictor of adverse outcomes, with multiple studies reporting a graded association between magnitude of proteinuria and risk of kidney failure.[4,7-9,52-54] The lowest risk for kidney failure is observed in those with a urine protein less than 0.5 g per day, in whom risk for kidney failure at 10 years from diagnosis is as low as approximately 5%.[52,53]

Kidney biopsy reports typically classify histologic features using the MEST-C score, which quantifies mesangial hypercellularity (M), endocapillary hypercellularity (E), segmental sclerosis (S), interstitial fibrosis/tubular atrophy (T), and crescents (C). The MEST variables, first described in 2009 based on a retrospective analysis of 265 adults and children with IgAN at a median follow-up of 5 years,[55] are useful to predict kidney outcomes.[6,56] In 2016, using a broader patient cohort, the original MEST working group determined that glomerular crescents were also predictive of worse outcome.[13] The KDIGO 2025 clinical practice guideline for IgAN recommends that

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all kidney biopsy reports for IgAN should provide a MEST-C score.[57] MEST-C histologic scores are also included in international IgAN prediction tools,[43,58] which should be used in all patients with IgAN and predict risk of 50% decline in eGFR or kidney failure over the next 5 to 7 years (eFigure in the Supplement). The original prediction tool, to be applied at the time of kidney biopsy, was derived and externally validated in multiethnic cohorts of adults with IgAN from Europe, North America, China, and Japan (derivation cohort n = 2781; validation cohort n = 1146).[43] A modified version, which incorporates similar data elements but can be applied 1 or 2 years after biopsy, was later developed and validated.[58] Although these prediction tools can refine prognosis, and inform patient counseling, they should not be used to make treatment decisions because there is insufficient evidence associating risk prediction with treatment response.[32,59]

Treatment of IgAN

According to the 2025 KDIGO clinical practice guidelines for IgAN, the treatment goal is to reduce the rate of eGFR decline to less than 1 mL/min/1.73 m^2 per year, which is best achieved by maintaining urine protein at less than 0.5 g per day or, preferably, less than 0.3 g per day.[13] Treatment of patients with IgAN at highest risk for progression, defined as those with proteinuria greater than 0.5 g per day (or an equivalent urine protein to creatinine ratio of >0.44 g/g[4]) includes behavioral modifications, antihypertensive medications to target a blood pressure less than 120/70 mm Hg, therapies to reduce the formation of IgA-containing immune complexes, decrease glomerular injury from IgA-containing immune complexes, and manage response to IgAN-induced nephron loss[13] (**Table 3** and **Table 4** and **Figure 3**).[60-69]

Behavioral Modification, Blood Pressure, and Lipid Control

Although evidence is largely extrapolated from general CKD populations, all patients with IgAN should be counseled about behavioral modifications to reduce progression of kidney and cardiovascular disease, including cessation of smoking and vaping, dietary sodium restriction (less than 2 g or 90 mmol per day), moderate exercise (at least 150 minutes per week as compatible with cardiovascular and physical tolerance), and weight loss, aiming for a body mass index target of 18.5 to 24.9.[13,70] Goal blood pressure is less than 120/70 mm Hg using standardized office blood pressure measurement,[13,71] with monitoring to avoid postural hypotension, electrolyte derangements, and acute decline in eGFR. The preferred first-line antihypertensive is a renin-angiotensin system inhibitor (RASi), ie, an angiotensin-converting enzyme inhibitor or angiotensin II receptor blocker.[13,70,71] In addition, adults 50 years or older with CKD not treated with dialysis or kidney transplant should receive a statin (1A recommendation if eGFR is less than 60 mL/min/1.73 m^2 and 1B if eGFR is greater than or equal to 60 mL/min/1.73 m^2); for adults younger than 50 years, statins are recommended for those with coronary artery disease, diabetes, prior ischemic stroke, or an estimated 10-year cardiovascular risk greater than 10% (2A recommendation).[70]

Management of IgAN-Specific Drivers of Nephron Loss

The KDIGO 2025 clinical practice guideline recommends that patients with IgAN and proteinuria greater than 0.5 g/d or equivalent (a urine protein to creatinine ratio greater than 0.44 g/g) receive

Table 2. Risk Factors for Kidney Disease Progression in IgA Nephropathy

Prognostic factor	HR (95% CI) for adverse kidney outcome
Variables included in the International IgAN Prediction Tool (at biopsy)[a]	
Age, per year	0.98 (0.98-0.99)[43,b,c]
eGFR, per mL/min/1.73 m^2	0.70 (0.66-0.75)[43,b,c]
Proteinuria, per g/d	1.69 (1.51-1.88)[43,b,c]
MAP, per mm Hg	1.01 (1.00-1.01)[43,b,c]
Race and ethnicity	
Japanese (reference: White race)	1.50 (1.07-2.11)[43,b,c]
Chinese (>3 y since biopsy) (reference: White race)	2.27 (1.73-2.96)[43,b,c]
Chinese (≤3 y since biopsy) (reference: White race)	0.67 (0.46-0.99)[43,b,c]
MEST histologic score	
M1 (vs M0)	1.34 (1.11-1.61)[43,b,c,d]
E1 (vs E0)	0.88 (0.69-1.13)[43,b,c,d]
S1 (vs S0)	1.15 (0.88-1.49)[43,b,c,d]
T1 (vs T0)	2.28 (1.84-2.82)[43,b,c,d]
T2 (vs T0)	3.04 (2.25-4.11)[43,b,c,d]
Use of RASi (vs not)	1.28 (1.04-1.58)[43,b,c,e]
Use of immunosuppression (vs not)	0.80 (0.57-1.11)[43,b,c,e]
Additional variables not included in the International IgAN Prediction Tool	
Demographic	
Male (reference: female)	0.90 (0.66-1.22)[44,c]
Clinical	
Persistent microhematuria (>5 RBC per HPF) (reference: ≤5 RBC per HPF)	2.84 (1.06-7.3)[45,c]
Hyperuricemia	RR, 2.50 (1.50-6.10)[46,f,g]
Obesity (reference: BMI <23 using Asian-specific classification or <25 using WHO classification)	OR, 2.43 (1.66-3.55)[47,h]
Tobacco smoking (reference: nonsmoking)	1.58 (1.02-2.46)[48,b,c]
Histologic	
Thrombotic microangiopathy	8.34 (1.66-41.96)[49,g]
Crescents >10% of glomeruli (reference: none)	1.80 (1.32-2.45)[50,b,c]
Crescents >25% of glomeruli (reference: none)	2.11 (1.47-3.02)[50,b,c]
C4d-positive staining	2.45 (1.30-4.64)[51,g]

Abbreviations: BMI, body mass index (calculated as weight in kilograms divided by height in meters squared); eGFR, estimated glomerular filtration rate; HPF, high-power field; HR, hazard ratio; IgAN, IgA nephropathy; MAP, mean arterial pressure; MEST-C, mesangial hypercellularity [M], endocapillary hypercellularity [E], segmental glomerulosclerosis [S], interstitial fibrosis/tubular atrophy [T], and crescents [C]; OR, odds ratio; RASi, renin-angiotensin system inhibitor; RBC, red blood cell; RR, relative risk; WHO, World Health Organization.

[a] The International IgA Nephropathy Prediction Tool[43] is recommended in the 2025 Kidney Disease: Improving Global Outcomes guideline for management of IgAN to predict risk for 50% decline in eGFR or kidney failure (defined as eGFR <15 mL/min/1.73 m^2, dialysis, or kidney transplant) over the next 5 to 7 years.[43]

[b] Decline in eGFR by at least 50%.

[c] Kidney failure defined as eGFR <15 mL/min/1.73 m^2, dialysis, or transplant.

[d] Limited model.

[e] Full model with race.

[f] Doubling of serum creatinine level.

[g] Initiation of kidney replacement therapy.

[h] Adverse kidney outcomes including development of kidney failure, a decline in eGFR by at least 30%, composite dialysis or death, occurrence of chronic kidney disease stage 3 and above defined as eGFR, calculated by abbreviated Modified Diet in Renal Diseases formula, below 60 mL/min/1.73 m^2.

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Table 3. Summary of Treatments to Reduce Immune-Complex Formation and Immune-Mediated Glomerular Injury in IgAN[a]

Drug	KDIGO recommended approach in IgAN	Summary of efficacy in IgAN	Adverse effects and cautions
Enteric targeted-release budesonide (Nefecon) 16 mg/d of budesonide for 9 mo then tapered over 2 wk	Recommendation 1.4.3.1 suggests treatment with a 9-mo course of budesonide for patients at risk of progressive loss of kidney function with IgAN (2B) Practice point 1.4.3.1: a 9-mo treatment course of budesonide, may not result in a sustained clinical response in terms of proteinuria reduction or eGFR stabilization	NefIgArd[60]: n = 364 Primary efficacy end point (time-weighted mean of eGFR over 2 y): beneficial effect with budesonide; difference of 5.05 mL/min/1.73 m^2 (95% CI, 3.24-7.38); $P < .0001$	Budesonide vs control: Peripheral edema, 31/182 (17%) vs 7/182 (4%) Hypertension in 31/182 (17%) vs 6/182 (3%) Muscle spasms in 22/182 (12%) vs 7/182 (4%) Acne in 20/182 (11%) vs 2/182 (1%)
Oral methlyprednisolone Reduced-dose protocol: methylprednisolone 0.4 mg/kg/d (maximum of 32 mg/d) for 2 mo, tapering by 4 mg/d each month for 6-9 mo Equivalent prednisone dose: 0.5 mg/kg/d, maximum 40 mg/d, tapering after 2 mo for a total o f 6-9 mo	Recommendation 1.4.3.2 suggests patients at risk of progressive loss of kidney function with IgAN be treated with a reduced-dose systemic glucocorticoid regimen combined with antimicrobial prophylaxis in settings where targeted-release budesonide is not available (2B) Practice point 1.4.3.2: (edited for brevity): Methylprednisolone (or equivalent) 0.4 mg/kg per day (maximum: 32 mg/d) for 2 mo, then taper by 4 mg/d each month for a total of 6-9 mo The conversion of methylprednisolone to prednisone or prednisolone is 1 mg methylprednisolone to 1.25 mg of prednisone or prednisolone	TESTING[61]: n = 503 Composite outcome of 40% eGFR decline, kidney failure, or death due to kidney disease: 74 events (28.8%) in the methylprednisolone vs 106 (43.1%) in the placebo group; HR, 0.53 (95% CI, 0.39-0.72) Serious adverse events: Severe infection requiring hospitalization: 17 events (7%) in methylprednisolone (12 in full-dose group and 5 in reduced-dose group) vs 3 (1%) in placebo group Gastrointestinal bleeding requiring hospitalization: 3 events in methylprednisolone (all in full-dose group) vs 1 in placebo group Clinically evident fracture or osteonecrosis: 3 events in methylprednisolone (all in full-dose group) vs 0 events in placebo group New-onset diabetes: 2 events in methylprednisolone group (both in reduced-dose group) vs 0 in placebo group	Practice point 1.4.3.2 (edited for brevity): Treatment with systemic glucocorticoids should incorporate antimicrobial prophylaxis against *Pneumocystis jirovecii* and antiviral prophylaxis in hepatitis B carriers, along with gastroprotection and bone protection according to local guidelines Practice Point 1.4.3.3 (edited for brevity): The following patient characteristics are likely to increase the risks of systemic glucocorticoid-related toxicity: eGFR <30 mL/min/1.73 m^2 Diabetes and prediabetes Obesity Latent infections (eg, viral hepatitis, tuberculosis) Active peptic ulceration Uncontrolled psychiatric illness Osteoporosis Cataracts
Mycophenolate mofetil (MMF) MMF 1.5 g/d for 12 mo, tapered to a maintenance dose of 0.75 to 1.0 g per day for at least 6 mo	Practice point 1.4.3.4: Chinese patients: in those patients in whom glucocorticoids are being considered, MMF may be used as a glucocorticoid-sparing agent Non-Chinese patients: there is insufficient evidence to support the use of MMF	MAIN[62]: n = 170 Composite outcome (doubling of serum creatinine, ESKD, or death due to kidney or cardiovascular cause): 6 (7.1%) in MMF group and 18 (21.2%) in the optimized supportive care group; adjusted HR, 0.23 (95% CI, 0.09-0.63); $P = .008$	Serious adverse events: 4/85 (4.7%) in MMF group vs 1/85 (1.2%) in control group: Infections: 17% vs 11% Gastrointestinal symptoms: 12% vs 0% Elevated serum transaminases: 6% vs 1% Leukopenia: 1% vs 0% Anemia: 8% vs 9%
Hydroxychloroquine 400 mg/d if eGFR >60 mL/min/1.73 m^2, 300 mg/d if eGFR 45-59 mL/min/1.73 m^2, and 200 mg/d if eGFR 30-44 mL/min/1.73 m^2	Practice point 1.4.3.4: Chinese patients: in those who remain at high risk for progression in spite of optimized supportive care Non-Chinese patients: there is insufficient evidence to support the use of hydroxychloroquine	Liu et al,[63] 2019: n = 60 Outcome: percentage change in proteinuria from baseline to 6 mo: 48.4% (IQR, −64.2% to −30.5%) in hydroxychloroquine vs 10.0% (IQR, −38.7% to 30.6%) in the placebo group; $P < .001$	No serious adverse events 23 Adverse events: eGFR reduction: 2/30 (6.7%) in hydroxychloroquine vs 0/30 (0%) in control group Nausea: 3% vs 0% Palpitations: 3% vs 0% Eye swelling/pain: 0% vs 3% Need for annual retinal screening

(continued)

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Table 3. Summary of Treatments to Reduce Immune-Complex Formation and Immune-Mediated Glomerular Injury in IgAN[a] (continued)

Drug	KDIGO recommended approach in IgAN	Summary of efficacy in IgAN	Adverse effects and cautions
Factor B inhibitor Iptacopan 200 mg twice/d	Trial published after 2025 KDIGO Clinical Practice Guidelines drafted; not included in current management algorithm Mentioned in section 1.6 Horizon scanning for future new drug approvals and updates to the guideline	APPLAUSE-IgAN[64] n = 443; 250 in interim analysis Interim primary outcome Adjusted geometric mean 24-h uPCR was 0.56 g/g in the iptacopan and 0.91 g/g in the placebo group (geometric mean ratio, 0.62 [95% CI, 0.51-0.74]; 2-sided $P < .001$) The adjusted geometric mean uPCR was 38.3% lower in the iptacopan vs placebo group (95% CI, 26.0% to 48.6%); 2-sided $P < .001$	Vaccinations against *Neisseria meningitidis*, *Streptococcus pneumoniae*, and *Haemophilus influenzae* type B must be completed prior to therapy Upper respiratory tract infection in 20/222 (9%) in iptacopan group vs 16/221 (7.2%) in placebo group in interim analysis

Abbreviations: APPLAUSE-IgAN, Targeting the Alternative Complement Pathway with Iptacopan to Treat IgA Nephropathy; eGFR, estimated glomerular filtration rate; ESKD, end-stage kidney disease; FDA, US Food and Drug Administration; HR, hazard ratio; IgAN, IgA nephropathy; KDIGO, Kidney Disease: Improving Global Outcomes; MAIN, Effect of Mycophenolate Mofetil on Renal Outcomes in Advanced Immunoglobulin A Nephropathy; NefigArd, Nefecon in Patients with Primary IgA Nephropathy at Risk of Progressing to End-Stage Renal Disease; TESTING, Therapeutic Evaluation of Steroids in IgA Nephropathy Global Study; uPCR, urine protein to creatinine ratio.

[a] Treatments are either recommended in the KDIGO 2025 Clinical Practice Guideline for Management of IgAN or recently approved by the FDA for treatment of IgAN but not (yet) included in guidelines.

treatment to prevent or reduce IgA-containing immune complex formation and IgA-containing immune complex-mediated glomerular injury.

The KDIGO 2025 clinical practice guideline recommends treatment with a 9-month course of targeted-release budesonide for patients with IgAN who are at risk of progressive loss of kidney function, defined as having proteinuria greater than 0.5 g per day or equivalent.[13] Targeted-release budesonide delivers glucocorticoid locally to the distal ileum, where B cells in Peyer patches are a key source of Gd-IgA1 production,[72-74] thereby reducing production of Gd-IgA1, which is the first step in IgA pathogenesis. A phase 3 RCT randomized 364 patients with IgAN, eGFR of 35 to 90 mL/min/1.73 m^2, and proteinuria greater than or equal to 1 g per day despite optimized supportive care to receive 9 months of targeted-release budesonide, 16 mg/d, or placebo.[75] After 2 years, change in eGFR from baseline was −6.11 (95% CI, −8.04 to −4.11) mL/min/1.73 m^2 in the targeted-release budesonide group and −12.00 (95% CI, −13.76 to −10.15) mL/min/1.73 m^2 in the placebo group, corresponding to a difference in the 2-year total eGFR slope of 2.95 (95% CI, 1.67-4.58) mL/min/1.73 m^2 per year in favor of targeted-release budesonide (P <.001).[60] However, targeted-release budesonide was associated with more corticosteroid-related adverse events such edema, hypertension, and acne[60] (Table 3).

The KDIGO 2025 guidelines recommend systemic glucocorticoids only when targeted-release budesonide is not available. The strongest evidence for their use comes from a multicenter double-blind RCT evaluating oral methylprednisolone, 0.6 to 0.8 mg/kg/d, vs placebo in 504 participants with IgAN and proteinuria greater than 1 g per day receiving optimized supportive care.[76] Although the original trial was stopped early due to 2 deaths in the treatment group, it was restarted with a modified treatment protocol using lower methylprednisolone dosing (0.4 mg/kg/d) and concurrent trimethoprim-sulfamethoxazole prophylaxis.[61] Over a mean of 4.2 years of follow-up, the primary outcome (composite of 40% decline in eGFR, kidney failure [dialysis, transplant], or death due to kidney disease) occurred in 74 participants (28.8%) in the methylprednisolone group compared with 106 (43.1%) in the placebo group (HR, 0.53 [95% CI, 0.39-0.72]; $P < .001$; absolute annual event rate difference, −4.8% [95% CI, −8.0% to −1.6%] per year).[61] However, higher rates of adverse events, particularly infections, were observed in the methylprednisolone group (Table 3).

Although not included in the KDIGO 2025 guideline for IgAN, iptacopan, an oral complement factor B inhibitor that targets glomerular inflammation ("hit" 4 in IgAN pathogenesis), has recently received US Food and Drug Administration (FDA) accelerated approval for the treatment of IgAN. In a phase 3 RCT of 443 patients with IgAN, proteinuria was reduced among those assigned to receive oral iptacopan (200 mg twice daily) compared with placebo. At 36 weeks, the adjusted geometric mean 24-hour urine protein to creatinine ratio was 38.3% (95% CI, 26.0%-48.6%) lower with iptacopan vs placebo (mean urine protein to creatinine ratio, 0.56 vs 0.91 g/g).[64] Results regarding the effect on eGFR at 2 years are awaited, which will determine whether iptacopan receives full FDA approval for IgAN.

Management of IgAN-Induced Nephron Loss

The KDIGO 2025 guideline[13] recommends that patients with IgAN should be treated with therapies to reduce glomerular hyperfiltration

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Table 4. Summary of Treatments to Manage the Consequences of IgAN-Induced Nephron Loss[a]

Drug	KDIGO recommended approach in IgAN	Summary of efficacy in IgAN	Adverse effects and cautions
RASi Ramipril 2.5-10 mg/d Perindopril 4-16 mg/d Losartan 25-100 mg/d Candesartan 8-32 mg/d	Recommendation 1.4.4.1: All patients with IgAN should be treated with an optimized maximally tolerated dose of either ACEi or an ARB (1B)	Cochrane review of nonimmunosuppressive treatment[65] Proteinuria: RASi vs placebo or no treatment (3 studies; n = 199): mean difference (MD) in proteinuria, −0.71 g/24 h (95% CI, −1.04 to −0.39) RASi vs symptomatic treatment (3 studies, n = 168): MD, −1.16 g/24 h (95% CI, −1.52 to −0.81) Creatinine clearance: RASi vs symptomatic treatment (2 studies, n = 127): MD, 23.26 mL/min (95% CI, 10.40-36.12) Remission, kidney failure, or doubling of serum creatinine: no significant between-group differences	Hyperkalaemia Acute dip in eGFR Cough (ACEi) Angioedema Practice Point 1.4.4.2: contraindications to this include low blood pressure, bilateral renal artery stenosis, or hyperkalemia
SGLT2i Dapagliflozin 10 mg/d Empagliflozin 10 mg/d	Recommendation 1.4.4.3: Patients at risk of progressive loss of kidney function with IgAN should be treated with an SGLT2i (2B)	DAPA-CKD[66]: n = 270 with IgAN Primary composite end point (sustained eGFR decline of at least 50%, kidney failure, or death from a kidney or cardiovascular cause): 6 (4%) in dapagliflozin vs 20 (15%) in placebo group: HR, 0.29 (95% CI, 0.12-0.73); *P* = .005 EMPA-KIDNEY[67]: n = 817 with IgAN Primary composite outcome (kidney failure, sustained eGFR<10 mL/min/1.73 m^2; sustained eGFR decline of at least 40%; or death from kidney failure or cardiovascular death): 51/413 (12.3%) in the empagliflozin group vs 67/404 (16.6%) in the placebo group; HR, 0.67 (95% CI, 0.46-0.97); eGFR decline was −2.33 vs −4.12 mL/min/1.73 m^2/y; absolute difference, 1.78 (95% CI, 1.13 to 2.44) mL/min/1.73 m^2/y	Urinary tract and mycotic genital infections Acute dip in eGFR Safety in combination with immunomodulatory or immunosuppressive therapies remains unproven There is uncertainty regarding the benefit of SGLT2i in younger patients with eGFR >60 mL/min/1.73 m^2 because DAPA-CKD and EMPA-KIDNEY cohorts were older with lower eGFR
DEARA Sparsentan 400 mg/d	Recommendation 1.4.4.2: Patients at risk of progressive loss of kidney function with IgAN should be treated with sparsentan (2B)	PROTECT[68]: n = 406 Primary end point: proteinuria reduction at 36 wk: −46.9% (95% CI, −53.4 to −39.5) sparsentan vs −5.9% (95% CI, −17.9 to 7.9) irbesartan Secondary end point: total eGFR slope to 110 wk: −2.9 vs −3.9 mL/min/1.73 m^2 per year (difference, 1.0 mL/min/1.73 m^2 per year [95% CI, −0.03 to 1.94]; *P* = .06)	Caution: Sparsentan is a DEARA and should not be prescribed together with a RASi TEAEs that occurred more frequently (≥5% difference) with sparsentan vs irbesartan included dizziness (15% vs 6%) and hypotension (13% vs 4%) Otherwise, as for RASi
Selective endothelin A receptor antagonist Atrasentan 0.75 mg/d	Trial published after 2025 KDIGO clinical practice guidelines drafted Approved by the FDA but not yet included in KDIGO guidelines	ALIGN[69]: n = 340; n = 270 in interim analysis Interim primary end point: mean uPCR reduction at 36 wk: −38.1% (95% CI −43.9 to −31.7) in the atrasentan group vs -3 .1% (95% CI,−12.4 to 7.3) in the placebo group	TEAEs that occurred more frequently (≥5% difference) with atrasentan vs placebo included anemia: 11/169 (6.5%) in atrasentan vs 2/170 (1.2%) in placebo group

Abbreviations: ACEi, angiotensin-converting enzyme inhibitors; ALIGN, A Phase 3, Randomized, Double-blind, Placebo-controlled Study of Atrasentan in Patients With IgA Nephropathy at Risk of Progressive Loss of Renal Function; ARB, angiotensin II receptor blockers; DAPA-CKD, Dapagliflozin and Prevention of Adverse Outcomes in Chronic Kidney Disease Study of Heart; DEARA, dual endothelin angiotensin receptor antagonist; eGFR, estimated glomerular filtration rate; EMPA-KIDNEY, Kidney Protection with Empagliflozin; FDA, US Food and Drug Administration; HR, hazard ratio; IgAN, IgA nephropathy; KDIGO, Kidney Disease: Improving Global Outcomes; MD, mean difference; PROTECT, Study of the Effect and Safety of Sparsentan in the Treatment of Patients With IgA Nephropathy; RASi, renin-angiotensin-aldosterone system inhibitors; SGLT2i, sodium-glucose cotransporter 2 inhibitors; TEAE, treatment-emergent adverse events; uPCR, urine protein to creatinine ratio.

[a] Treatments are either recommended in the KDIGO 2025 Clinical Practice Guideline for Management of IgAN or recently approved by the FDA for treatment of IgAN but not (yet) included in guidelines.

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Figure 3. Treatment of IgAN

Adult patient with IgA nephropathy (IgAN) at risk of progressive kidney function loss
Proteinuria ≥0.5 g/d while receiving or not receiving IgAN treatment

Goal is to reduce the rate of annual estimated glomerular filtration rate (eGFR) decline to <1 mL/min/1.73 m^2 and maintain urine protein excretion <0.5 g/d (ideally <0.3 g/d)

Consider simultaneous management of IgAN-specific drivers for nephron loss and generic responses to IgAN-induced nephron loss

Manage IgAN-specific drivers for nephron loss

Stop synthesis of pathogenic forms of IgA and IgA immune-complex formation
- Targeted release formulation budesonide, 16 mg/d for 9 mo
- BAFF/APRIL inhibitors *(under investigation)*

Stop IgA-mediated kidney injury and IgA immune complex-mediated kidney injury
- Targeted-release budesonide, 16 mg/d for 9 mo
- If budesonide not available, oral methylprednisone, 0.4 mg/kg/d, (maximum 32 mg/d) for 2 mo and weaning monthly by 4 mg/d for 6-9 mo
  or
  Prednisone, 0.5 mg/kg/d, (maximum 40 mg/d) for 2 mo and weaning monthly by 5 mg/d for 6-9 mo[a]
- Iptacopan, 200 mg twice daily *(recently licensed but not yet recommended by KDIGO)*
- Other complement therapeutics *(under investigation)*

Manage generic response to IgAN-induced nephron loss

Reduce glomerular hyperfiltration and impact of proteinuria on tubulointerstitium
and
Control blood pressure
- Lifestyle modifications: smoking and vaping cessation, dietary sodium restriction of <200 mg/d or 90 mmol/d, 150 min/wk of moderate exercise, and weight management
- ACE inhibitor or ARB[b] at maximum tolerated dose
- Sparsentan,[b] 200 mg/d increased to 400 mg/d as tolerated after 14 d
- Sodium-glucose cotransporter 2 inhibitor (SGLT2i)
- Atrasentan, 0.75 mg/d, as an alternative to sparsentan *(recently licensed but not yet recommended by KDIGO)*

- ***To manage cardiovascular risk****, begin statin therapy for patients with chronic kidney disease >50 y and for patients ≤50 y if 10-y cardiovascular risk >10% or coexisting cardiovascular disease*
- ***Treatment of variant forms of primary IgAN*** *(IgAN with nephrotic syndrome, IgAN with AKI, and IgAN with RGPN) are summarized in eTable 5 in the Supplement*
- ***Treatment of secondary IgAN*** *should focus on the specific disease process triggering secondary IgAN with chronic kidney disease management (ie, lifestyle modification, blood pressure control, ACE inhibitor or ARB with or without SGLT2i)*

AKI, acute kidney injury; APRIL, a proliferation-inducing ligand; BAFF, B cell–activating factor; KDIGO, Kidney Disease: Improving Global Outcomes; RGPN, rapidly progressive glomerulonephritis.

[a]Sparsentan should not be prescribed together with an angiotensin-converting enzyme (ACE) inhibitor or angiotensin II receptor blocker (ARB) because sparsentan already combines a renin-angiotensin system inhibitor (RASi) with an endothelin antagonist.

[b]Treatment with systemic glucocorticoids should incorporate antimicrobial prophylaxis against *Pneumocystis jirovecii* and antiviral prophylaxis in hepatitis B carriers, along with gastroprotection and bone protection according to national guidelines.

and pathways leading to tubulointerstitial fibrosis, including RASi, sparsentan (a dual endothelin and angiotensin II receptor antagonist), and sodium-glucose cotransporter 2 inhibitors (SGLT2i) (Table 4 and Figure 3).

The KDIGO 2025 guideline recommends treatment of all patients with IgAN with a maximally tolerated dose of an angiotensin-converting enzyme inhibitor or angiotensin II receptor blocker, except for those with contraindications such as low blood pressure, bilateral renal artery stenosis, or hyperkalemia.[13] RASi have shown benefit in small clinical trials of patients with IgAN.[77-80] In a meta-analysis of 3 of these studies (n = 199), RASi significantly reduced proteinuria compared with placebo (standardized mean difference, −0.71 g/d [95% CI, −1.04 to −0.39]),[65] although differences in eGFR were not statistically significant.

The KDIGO 2025 guideline recommended use of a dual endothelin and angiotensin II receptor antagonist (sparsentan) as an alternative to RASi. Sparsentan should not be used with RASi because it contains an angiotensin II antagonist with an endothelin antagonist. A phase 3, double-blind, multicenter RCT of 406 patients with IgAN reported proteinuria reduction at 110 weeks of 43% with sparsentan 400 mg daily vs 4% with irbesartan 300 mg daily (geometric least-squares mean ratio, 0.60 [95% CI, 0.50-0.72]).[68] Decline in eGFR (week 6 to 110) was also reduced with sparsentan, with a slope of −2.7 mL/min/1.73 m^2 per year vs −3.8 mL/min/1.73 m^2 per year with irbesartan (difference, 1.1 [95% CI, 0.1-2.1] mL/min/1.73 m^2 per year; P = .04).[68] No significant differences in treatment-related adverse events were observed.[68]

The KDIGO 2025 guideline also recommended SGLT2i for patients with IgAN and proteinuria greater than 0.5 g per day.[13] Although dedicated trials with SGLT2i have not been performed, post hoc analyses of patients with IgAN enrolled in 2 CKD trials have been reported.[66,67] The first study randomized 4304 patients with eGFR of 25 to 75 mL/min/1.73 m^2 and urinary albumin to creatinine ratio of 200 to 5000 mg/g to receive dapagliflozin, 10 mg/d, or placebo.[66] In the IgAN subgroup (n = 270), the primary composite end point (≥50% eGFR decline, kidney failure, or kidney or cardiovascular death) occurred in 4% (6 participants) of the dapagliflozin group vs 15% (20 participants) in the placebo group (HR, 0.29 [95% CI, 0.12-0.73]; P = .005; absolute risk difference, 10.7% [95% CI, −17.6% to −3.7%]).[66] Adverse events were similar between groups.[66] In the IgAN subgroup (n = 817) of the second trial, the primary composite outcome (kidney failure, sustained eGFR <10 mL/min/1.73 m^2, sustained eGFR decline of at least 40%, or death from kidney failure or cardiovascular death) was lower among patients randomized to receive the SGLT2i empagliflozin 10 mg/d (12.3% [51/413]) vs placebo (16.6% [67/404]) (HR, 0.67 [95% CI, 0.46-0.97]).[67]

Although not included in the KDIGO 2025 guidelines, atrasentan (a selective inhibitor of the endothelin type A receptor) received accelerated FDA approval for IgAN. Interim results of a phase 3 RCT of 270 patients with IgAN reported decreased proteinuria with atrasentan 0.75mg once daily compared with placebo at week 36 (mean urine protein to creatinine ratio decreased from 1450 to 882 mg/g with atrasentan vs 1484 to 1375 mg/g with placebo).[69] Evidence about effects of atrasentan on eGFR at 2 years is awaited,

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which will determine whether atrasentan receives full FDA approval for IgAN.

Management of Variant Forms of IgAN

Clinical characteristics, histologic features, and the KDIGO 2025 guideline treatment recommendations[13] for variant forms of primary IgAN (IgAN with nephrotic syndrome, IgAN with acute kidney injury, and IgAN with rapidly progressive glomerulonephritis) are summarized in eTable 6 in the Supplement.

Prognosis

More than half of patients diagnosed with primary IgAN develop kidney failure within 20 years of their diagnosis (Table 1). Among 2299 adults with biopsy-proven IgAN included in a national registry–based study in the UK, 50% developed the composite outcome of death, kidney failure, or an eGFR less than 15 mL/min/1.73 m^2 at a median (IQR) of 10.8 (10.0-12.0) years after diagnosis, at median age 49 years.[4] For patients diagnosed with IgAN between the ages of 30 and 39 years, the percentage estimated to develop kidney failure was 100% if eGFR decline was 3 mL/min/1.73 m^2 per year, 80% if eGFR decline was 2 mL/min/1.73 m^2 per year, 38% if eGFR decline was 1 mL/min/1.73 m^2 per year, and 19% if eGFR decline was 0.5 mL/min/1.73 m^2 per year.[4] In a US cohort of 655 adults with biopsy-proven IgAN enrolled in a Southern Californian health system, 179 of 655 (27%) experienced kidney failure (eGFR <15 mL/min/1.73 m^2, dialysis, or kidney transplant) at a median (IQR) of 2.6 (0.6-5.7) years and a median (IQR) age 48.8 (41.8-61.9) years.[9]

Although kidney transplant is recommended for most patients with IgAN who develop kidney failure, IgAN can recur following transplant. In a multicenter cohort of adults with biopsy-proven IgAN across Europe, North America, and South America, the cumulative incidence of IgAN recurrence after kidney transplant was 19% (95% CI, 12%-26%) at 10 years and 23% (95% CI, 14%-34%) at 15 years.[81] IgAN recurrence in the transplanted kidney was associated with an HR of 3.69 (95% CI, 2.04-6.66) for graft failure, and 10-year death-censored graft survival was 76% in patients with IgAN recurrence vs 89% in those without recurrence.[81]

In a Swedish population-based cohort study of 3622 adults with IgAN and 18 041 matched controls, IgAN was associated with a 1.53-fold increased risk of death, with an absolute excess mortality of 3.2 per 1000 person-years, and lowered median life expectancy by 6 years.[5]

Practical Considerations

Treatment options to prevent or reduce IgA-containing immune complex formation and IgA-containing immune complex–mediated glomerular injury currently include targeted-release budesonide or systemic glucocorticoids, but iptacopan has received accelerated approval by the FDA for treatment of IgAN. Whether or how to combine these therapies is currently unclear. In addition, head-to-head trials of recently licensed therapies for IgAN (targeted-release budesonide, sparsentan, iptacopan, and atrasentan) have not been conducted; therefore, relative efficacy and the preferred order or combination of these drugs is unknown. Moreover, the duration of therapy, and recommendations about repeat courses of treatment, also remain to be established. In addition, there is insufficient evidence to guide treatment in patient populations that have been excluded from IgAN clinical trials (ie, children and adolescents, patients with IgAN recurrence in a kidney transplant, patients with eGFR less than 30 mL/min/1.73 m^2, patients with proteinuria less than 0.75 to 1.0 g per day, and patients who are pregnant or planning to conceive). Additionally, recently licensed therapies for IgAN may not be accessible or affordable in all countries, limiting available treatment options.

Future Directions

Therapies under investigation in phase 2 and 3 clinical trials, some of which have reported interim results that could lead to accelerated FDA approval,[82,83] are shown in eTables 4 and 5 in the Supplement. Identification and validation of serum and urine biomarkers that could inform diagnosis, prognosis, and treatment monitoring in IgAN is another area of active research.[84-97]

Limitations

This review has several limitations. First, the quality of included studies was not formally evaluated. Second, the literature search was limited to English-language publications, and some relevant studies may have been missed. Third, this review focused on adults and did not include pediatric populations.

Conclusions

IgAN is the leading cause of immune-mediated glomerular disease worldwide. Patients with suspected IgAN and proteinuria greater than or equal to 0.5 g per day should undergo kidney biopsy. Treatment of IgAN includes behavioral modifications, blood pressure management, and therapies to decrease formation of IgA-containing immune complexes (eg, targeted-release budesonide), reduce immune complex–mediated glomerular injury (eg, systemic glucocorticoids, iptacopan), and manage IgAN-induced nephron loss (eg, RASi, sparsentan, atrasentan, SGLT2i).

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Submissions: We encourage authors to submit papers for consideration as a Review. Please contact Kristin Walter, MD, at kristin.walter@jamanetwork.org.

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